Secure Cloud Storage Architecture, Backups & Business Continuity-Disaster Recovery Implementation

1. Project Overview

This project involved the design and implementation of a secure, resilient storage infrastructure in Microsoft Azure. The goal was to implement a multi-tiered storage solution, incorporating Blob Storage and Azure Files, while enforcing strict network security, automated lifecycle management, and a robust Backup and Business continuity (BCDR) strategy using Recovery Services Vaults.

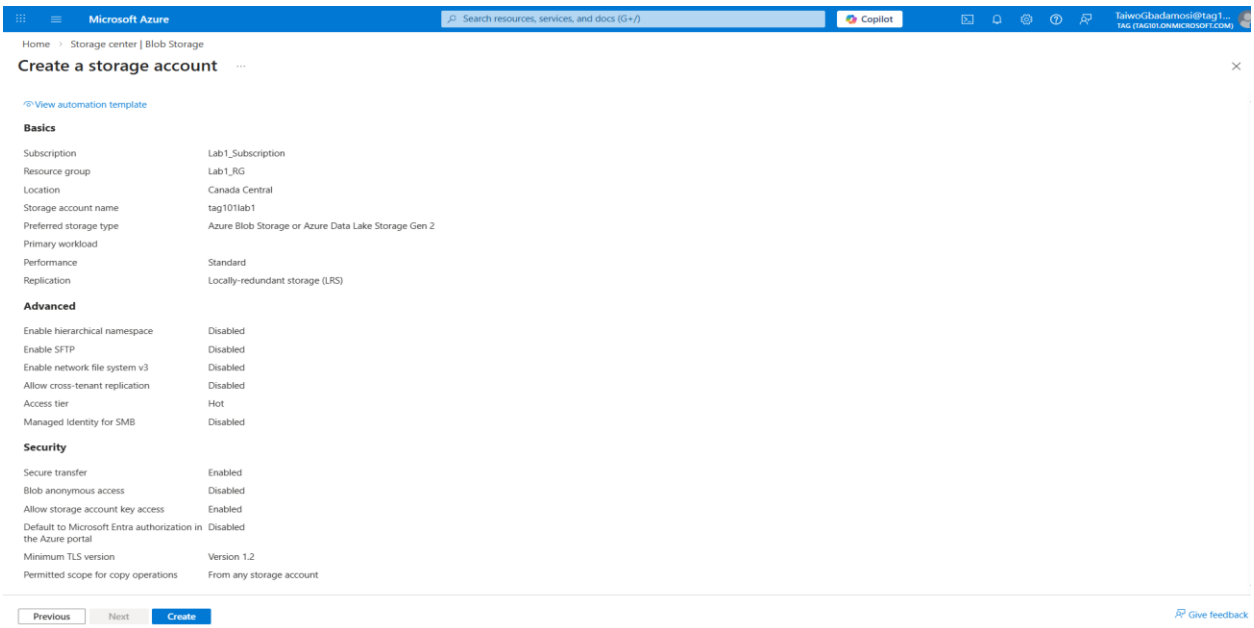
2. Tools & Methodologies

- **Storage Services:** Azure Blob Storage, Azure Files (SMB 3.0).
- **Security:** Network Firewalls (VNet Integration), Shared Access Signatures (SAS), RBAC.
- **Automation:** Lifecycle Management Policies.
- **Data Protection:** Recovery Services Vault (RSV), Azure Backup.
- **Disaster Recovery:** Azure Site Recovery (ASR), Replication Policies.
- **Compute/Access:** Azure Virtual Machines (Windows Server), RDP.

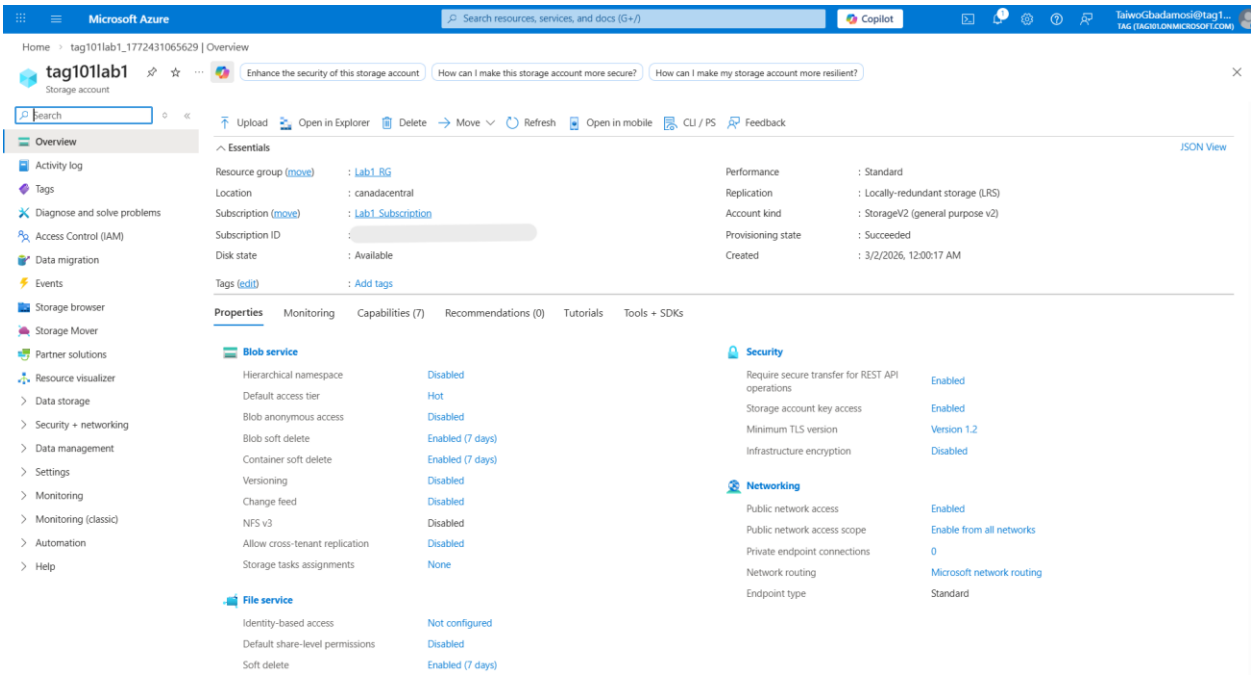
3. Technical Implementation Phases

Phase I: Secure Blob Storage & Lifecycle Management

- **Storage Account Provisioning:** Created a redundant storage account, while also enabling the "Soft Delete" settings in the account. This effectively adds an extra layer of protection against accidental deletion.

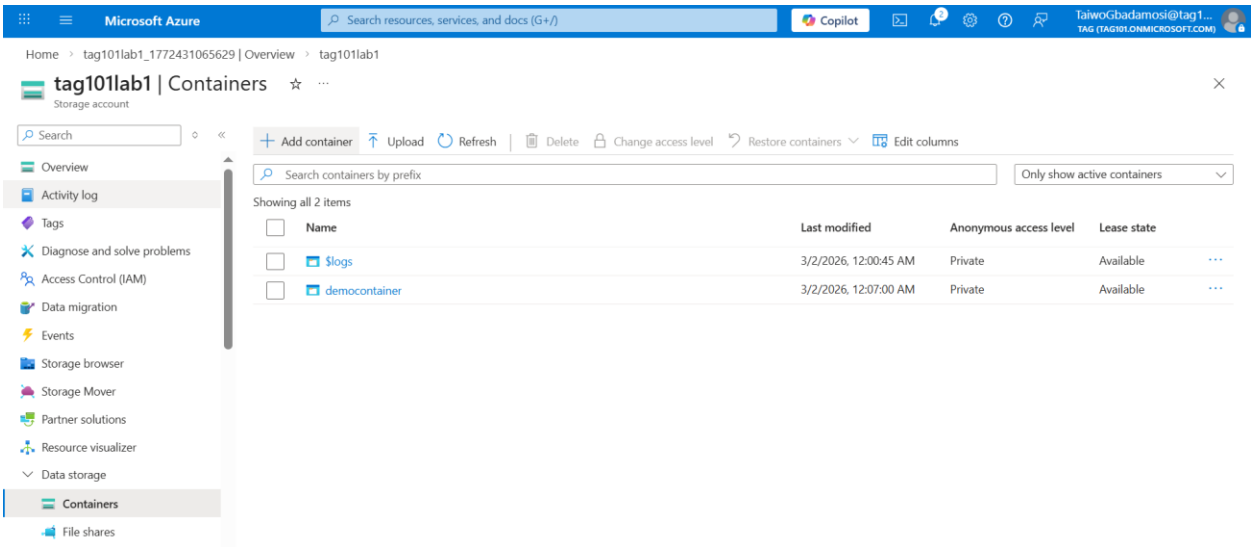


Azure storage account configuration

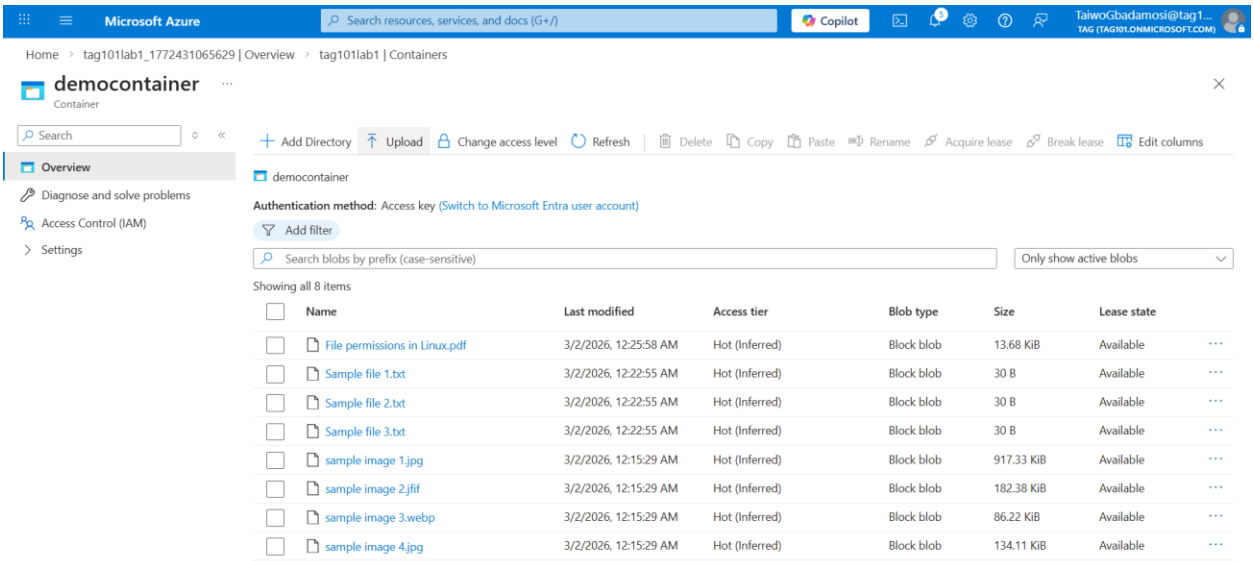


Azure storage account overview

- **Blob Container Provisioning:** Deployed a blob container within the storage account, and uploaded sample blobs to the container

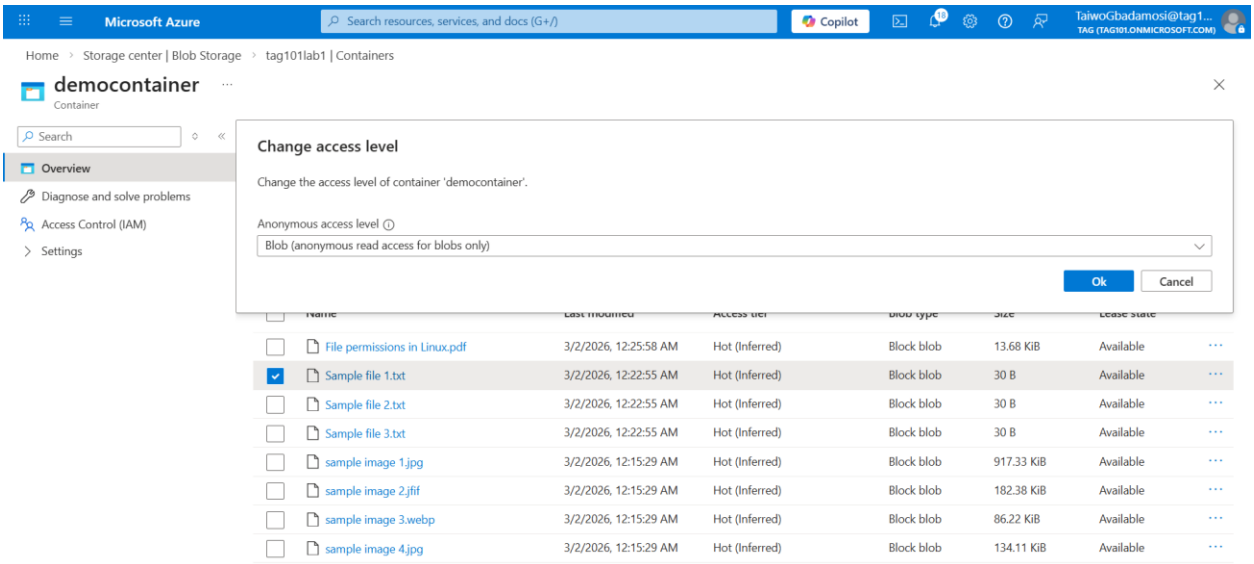


Azure container configured for blob storage

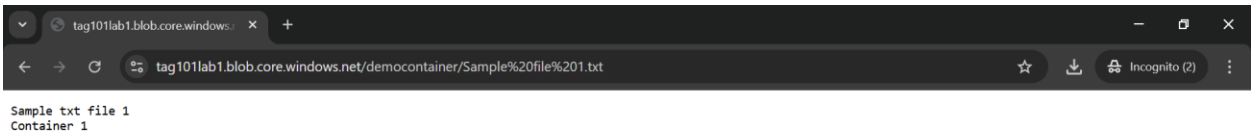


Sample blobs uploaded to the blob container

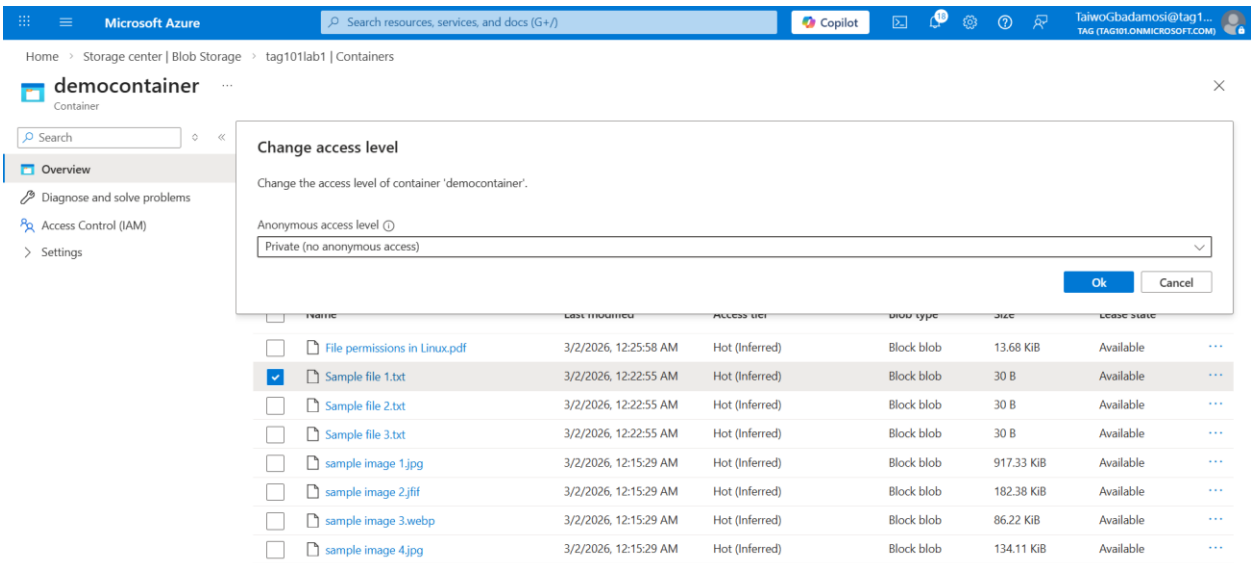
- **Unauthenticated Read access:** Restricted public read access via https on the internet to private access to enforce Security Hardening.



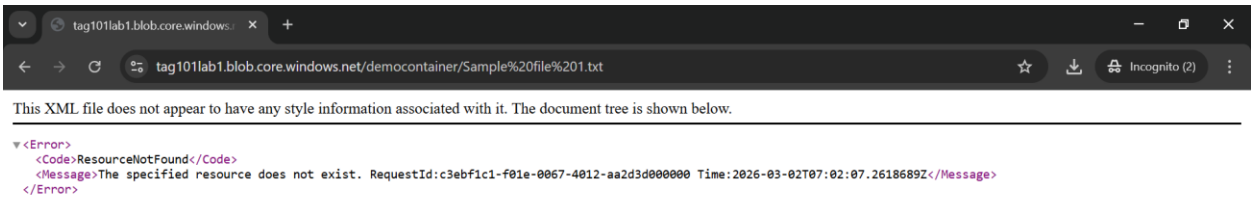
Default configuration allowing anonymous read access for the blob files



Blob file successfully accessed on the internet

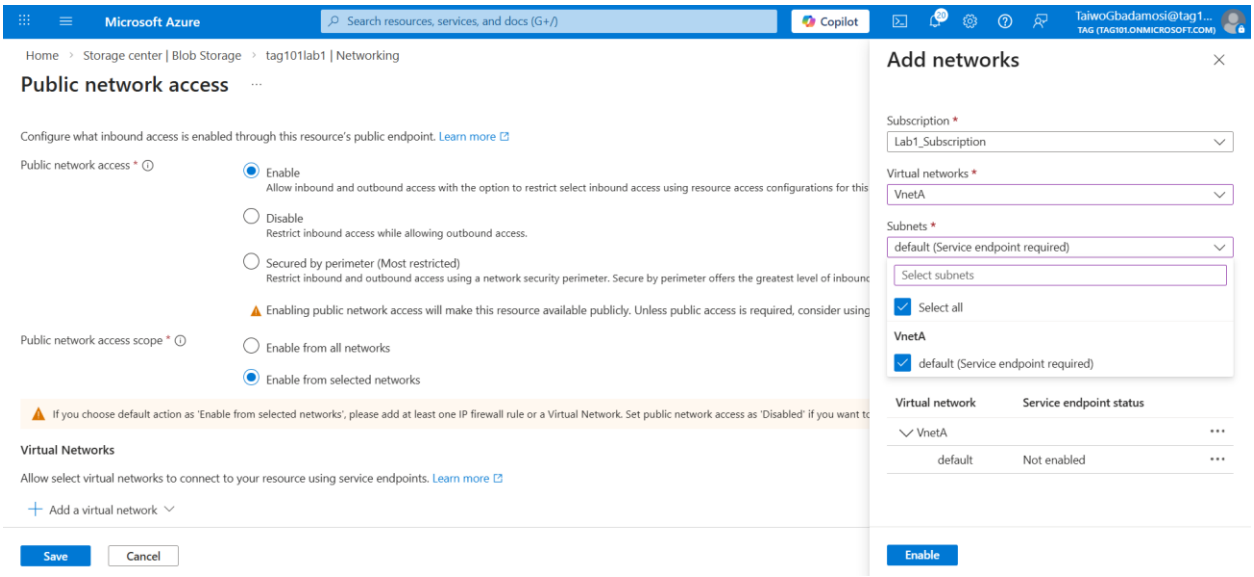


Configuring private access for access level to deny unauthorized https access



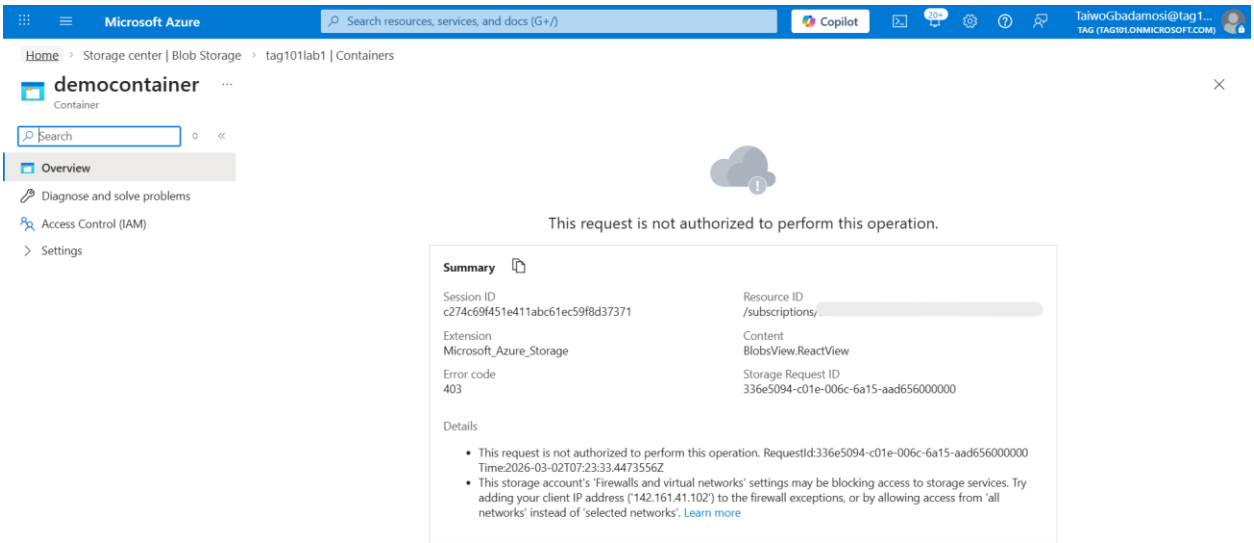
Access denied: unauthorized https access

- **Network Hardening:** Restricted access to "Selected Networks."



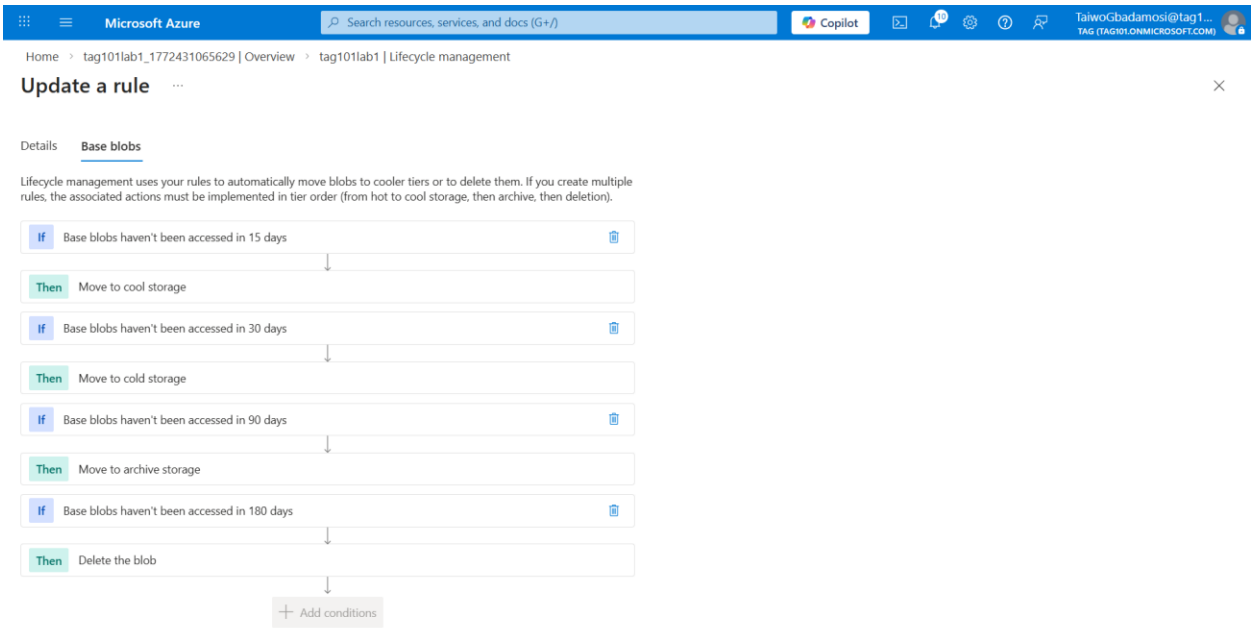
Configuring access scope for public network access to only select networks

- **Validation:** Attempted access from an unauthorized IP and access was correctly denied for unauthorized attempt to operate blob storage container.



Unauthorized access to blob container denied

- **Lifecycle Management:** Configured rules to automatically move blobs to Cool, Cold and Archive tiers after a set period to optimize costs.

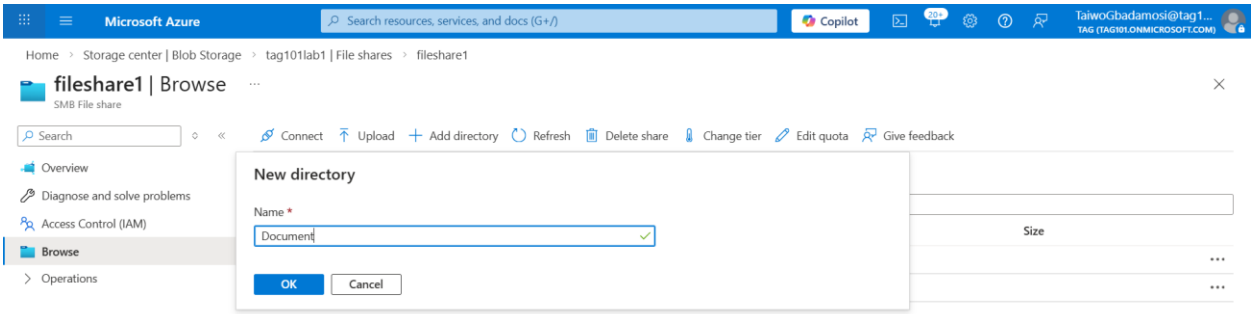


Lifecycle Management rule logic

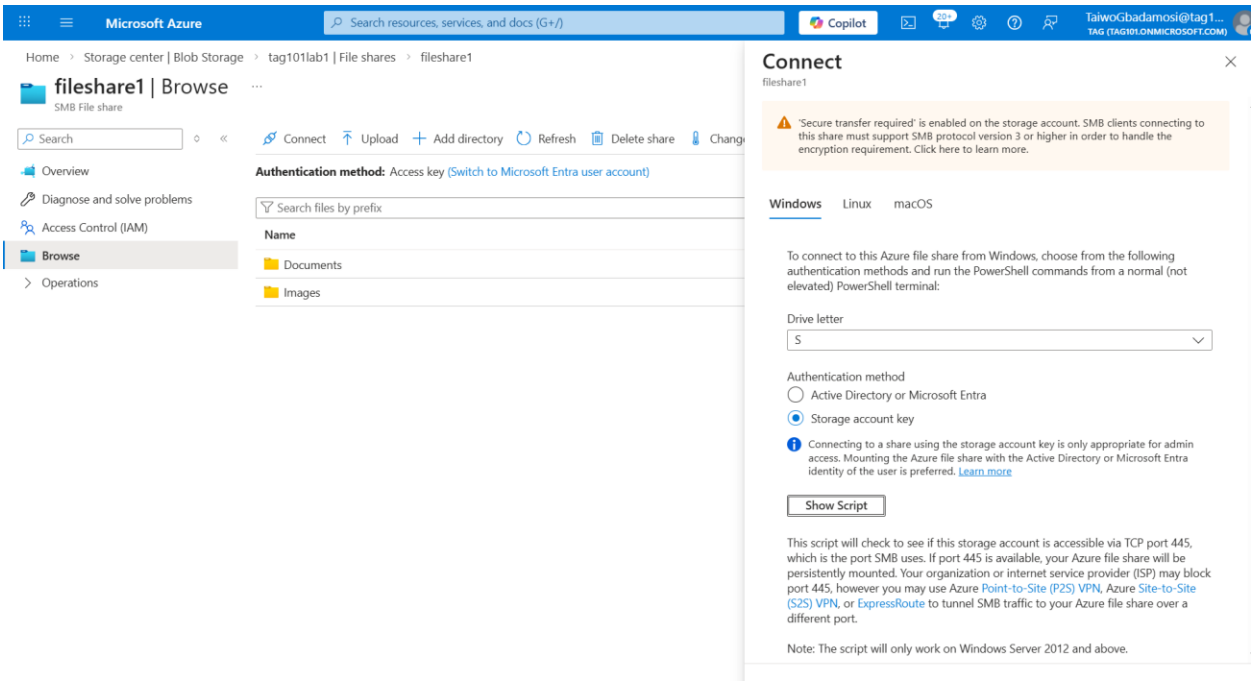
Phase II: Azure Files & VM Integration

- **File Share Creation:** Established a SMB-based file share directory.

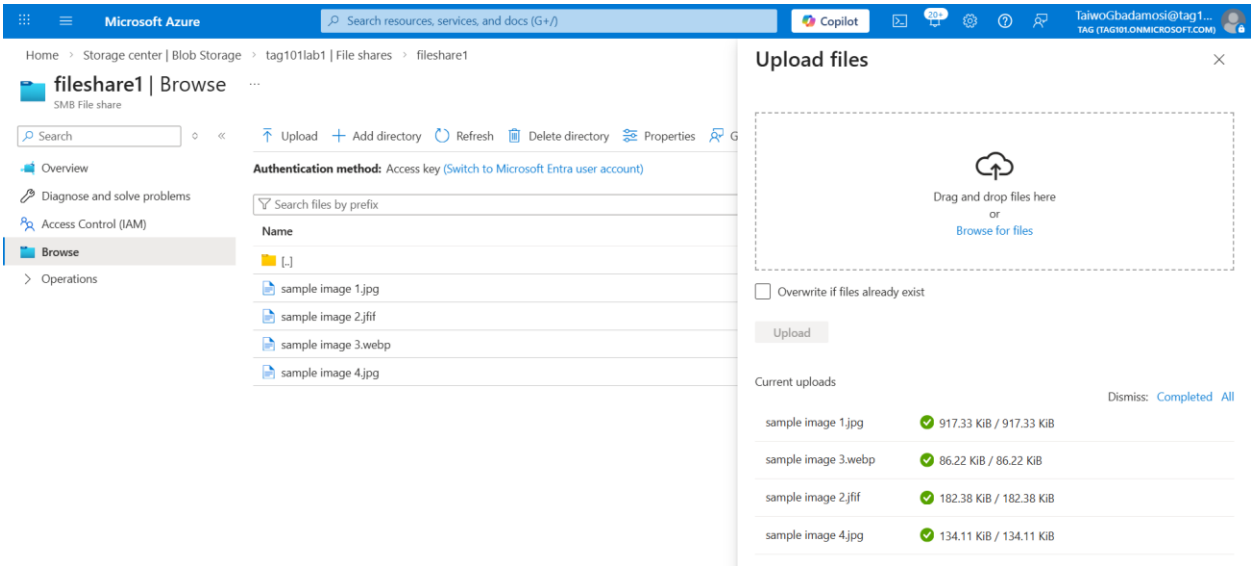
Configuration of new file share account



Adding new directories to file share

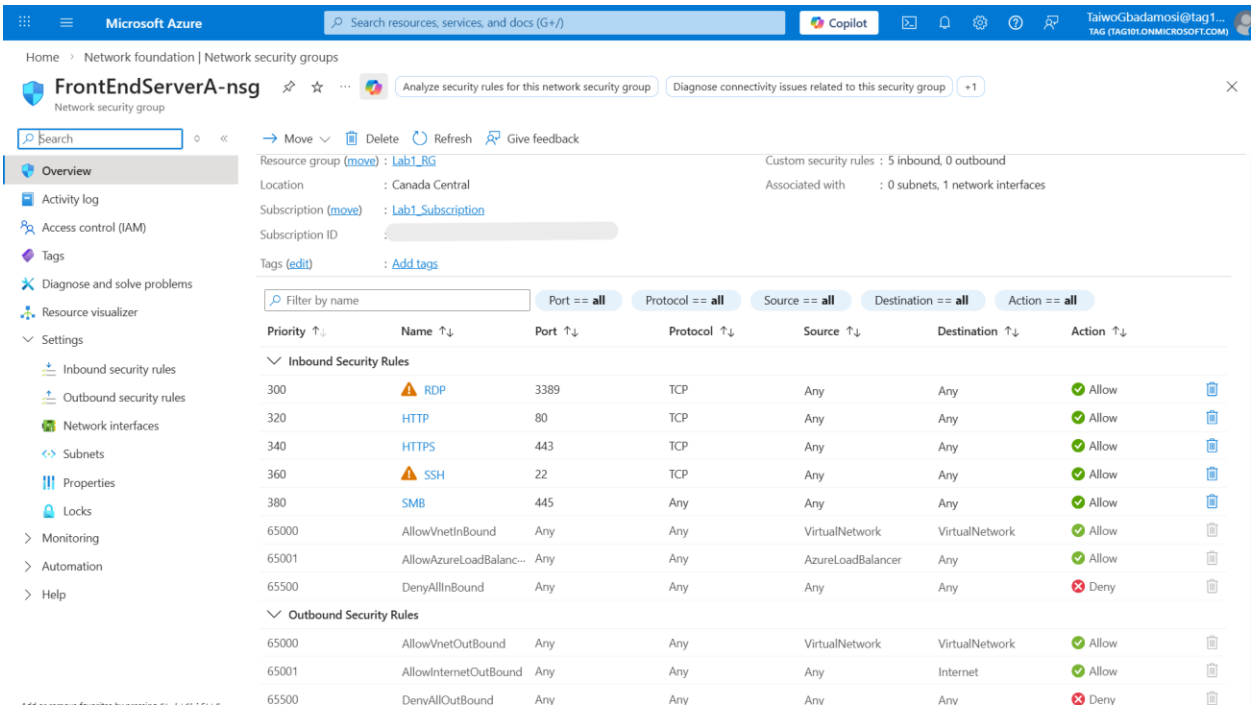


File shares showing populated directories generating script to be mapped as a network drive on vm



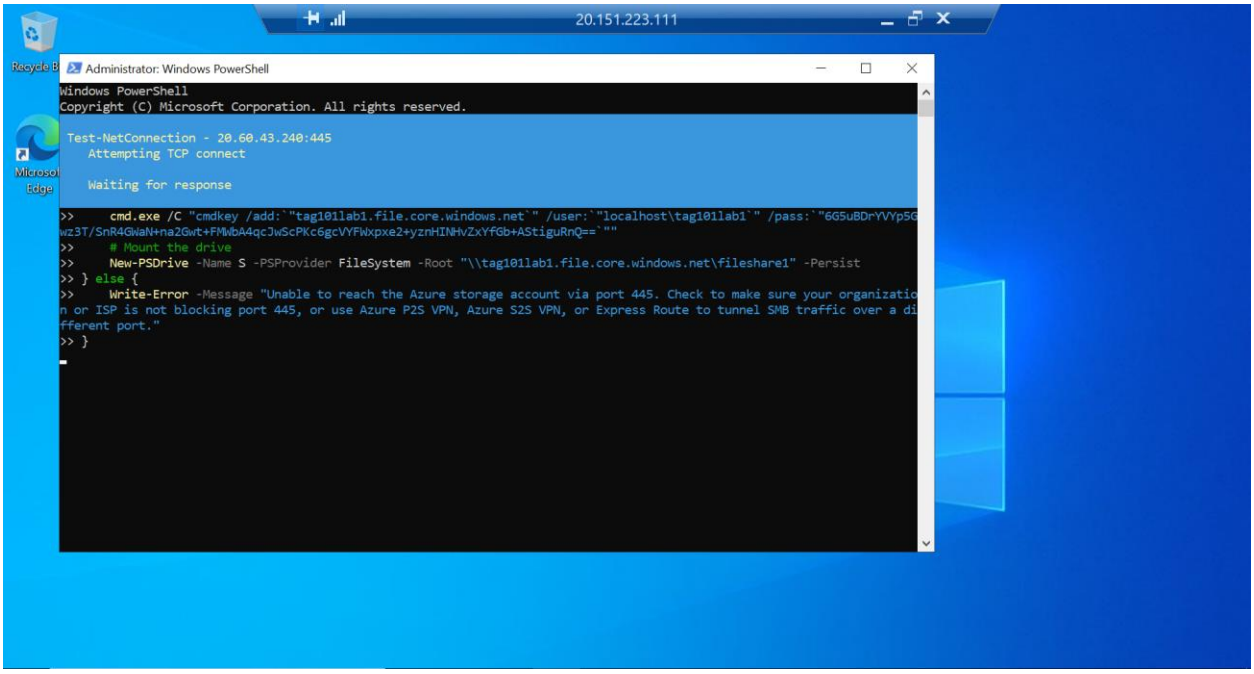
Uploading new files to directory in file share

- **Virtual Machine Deployment:** Created a Windows vm to act as the client.
- **Connectivity:** Configured a Network Security Group (NSG) rule to allow SMB (Port 445) traffic.



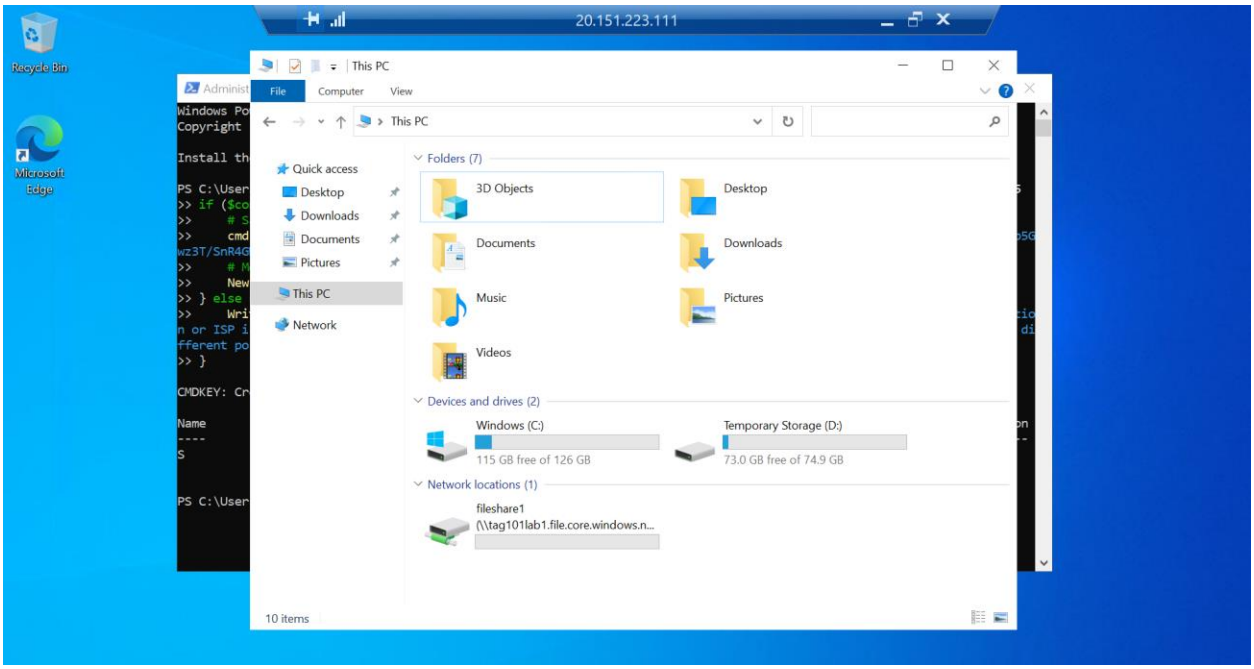
Confiuring SMB allow rule on virtual machine network interface network security group

- **Mounting:** Generated a PowerShell mounting script from the Azure Portal and executed it via RDP on the VM.

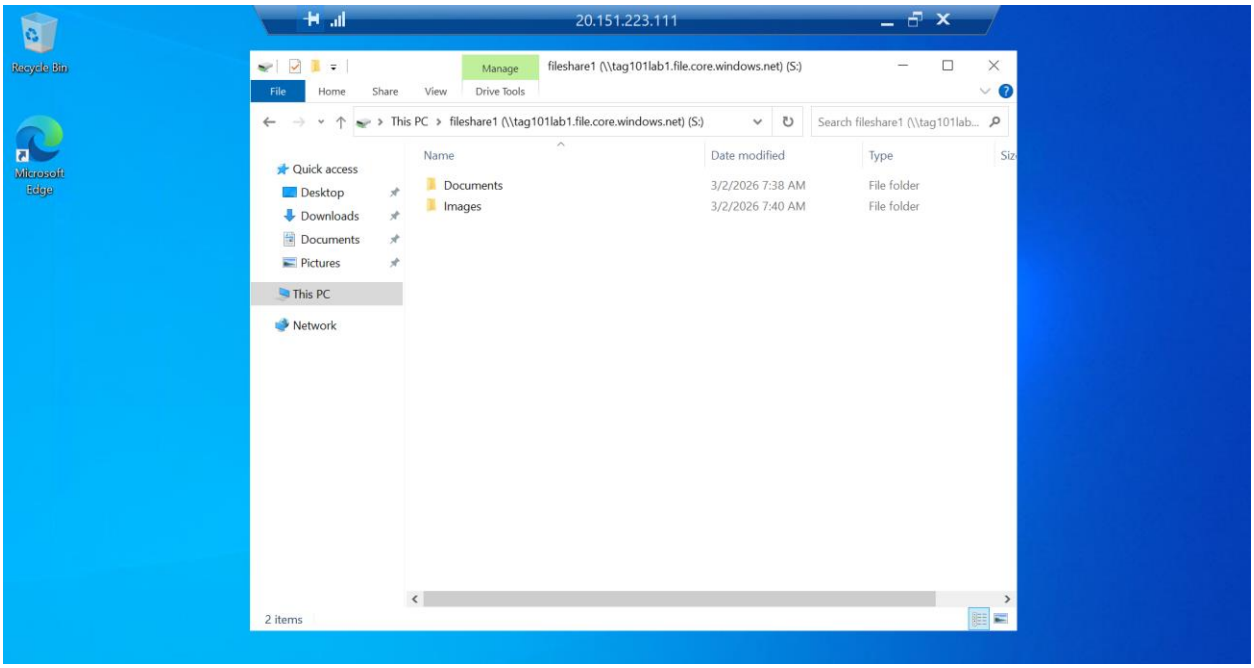


Powershell script execution

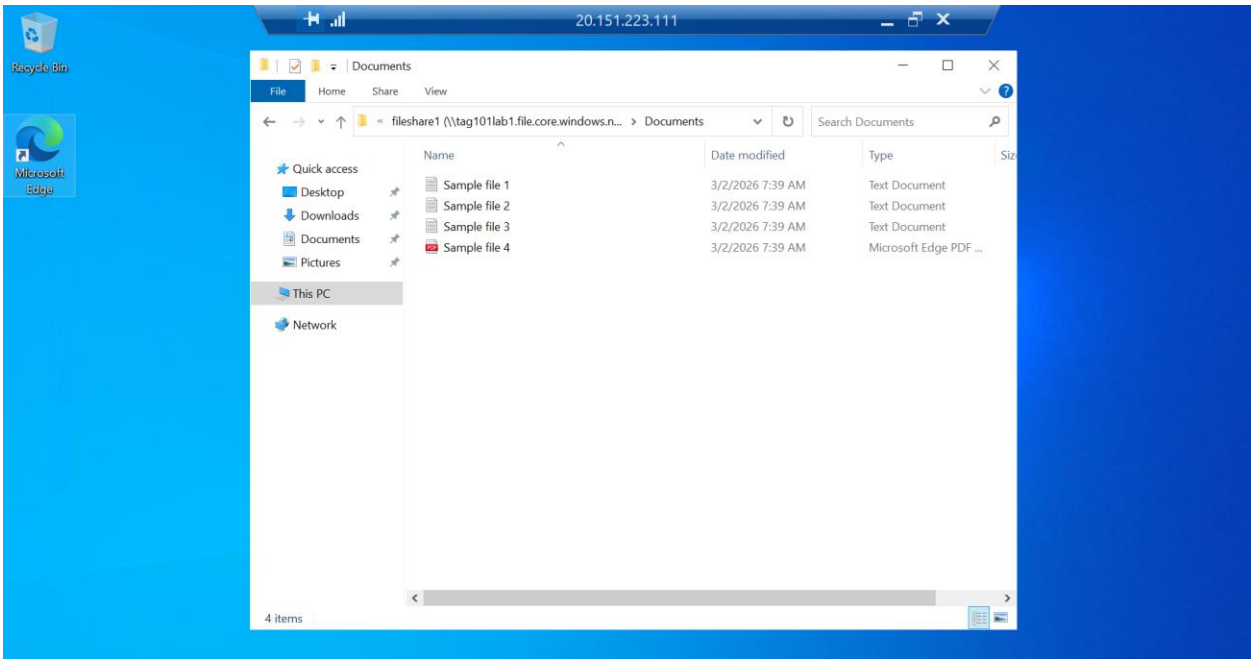
- **Verification:** Confirmed the File Share was mapped as a network drive (S: drive).



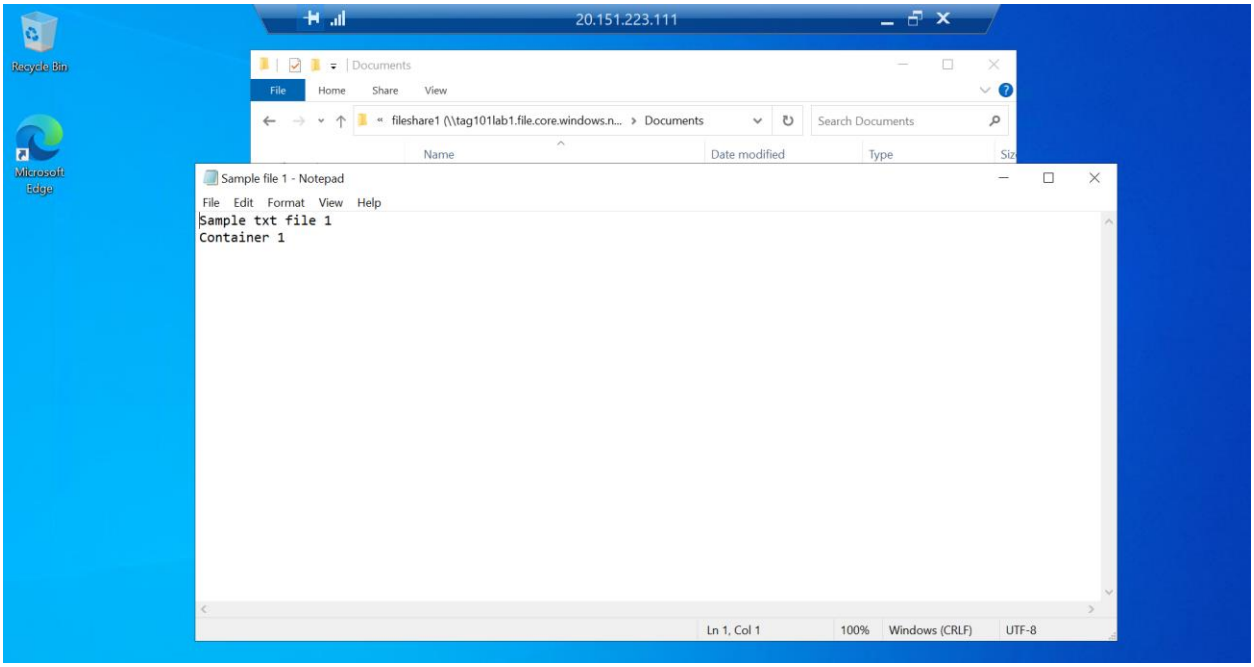
Successful SMB Mount: Azure File Share mapped as a Network Drive



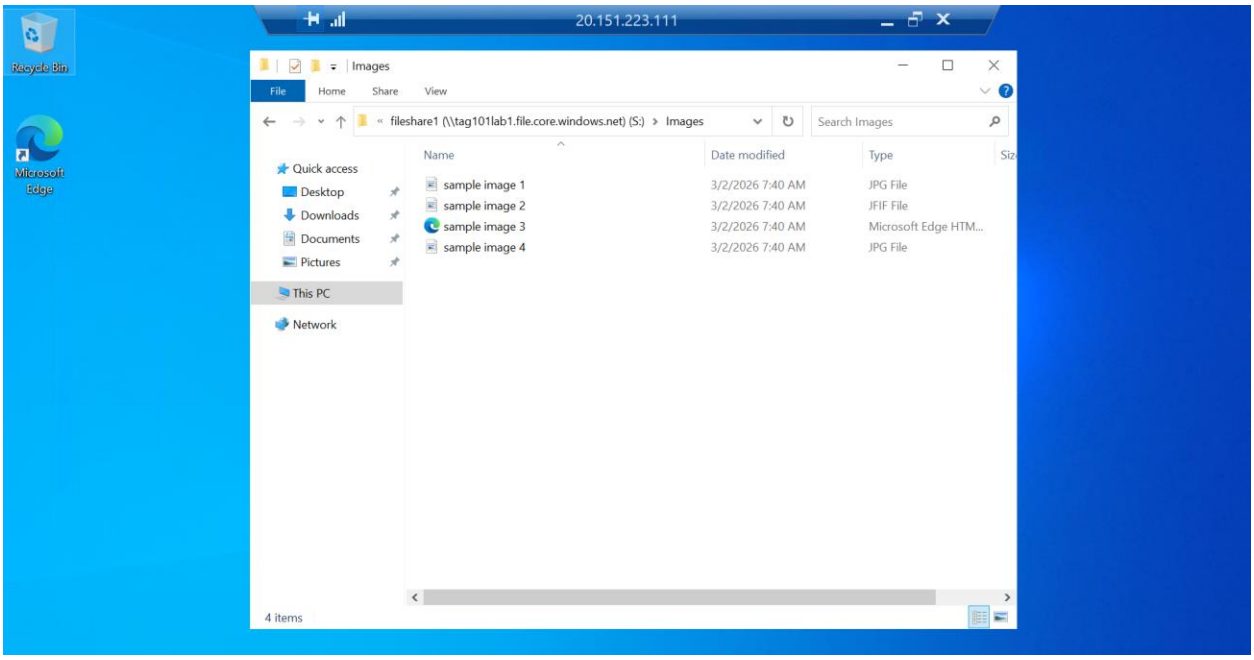
Accessing Azure File Share Subfolders via Mapped Drive



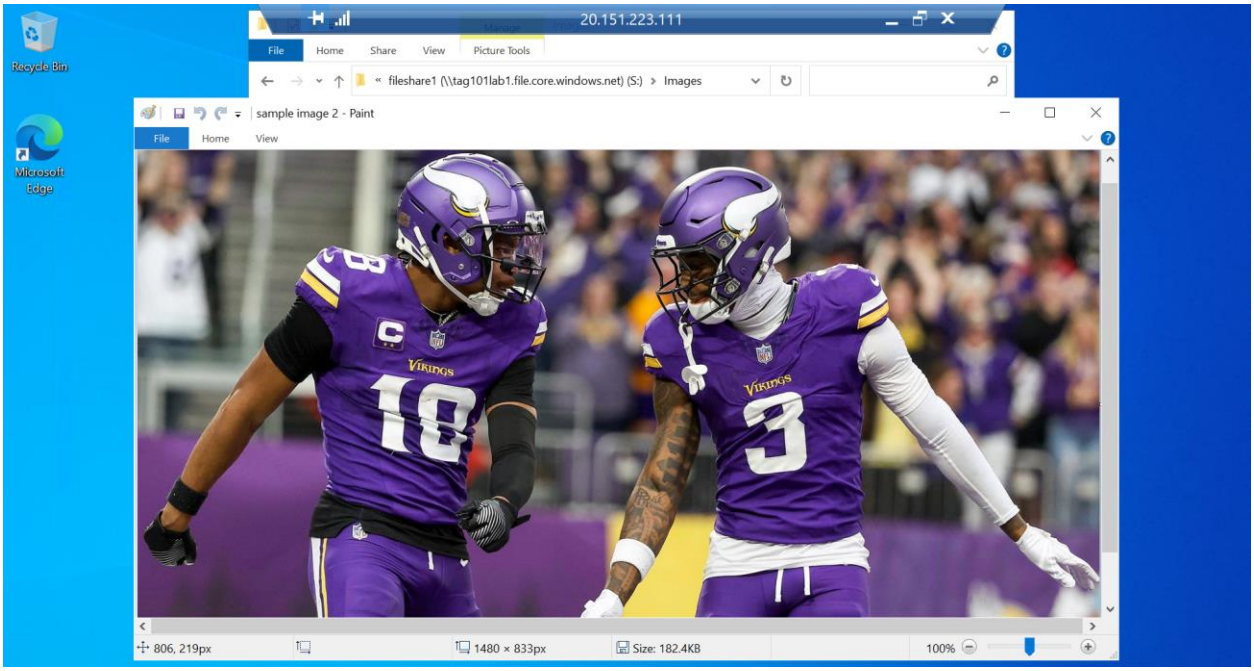
Accessing Azure File Share Subfolder "Documents" via Mapped Drive



Accessing Azure File Share file within subfolder "Documents" via Mapped Drive



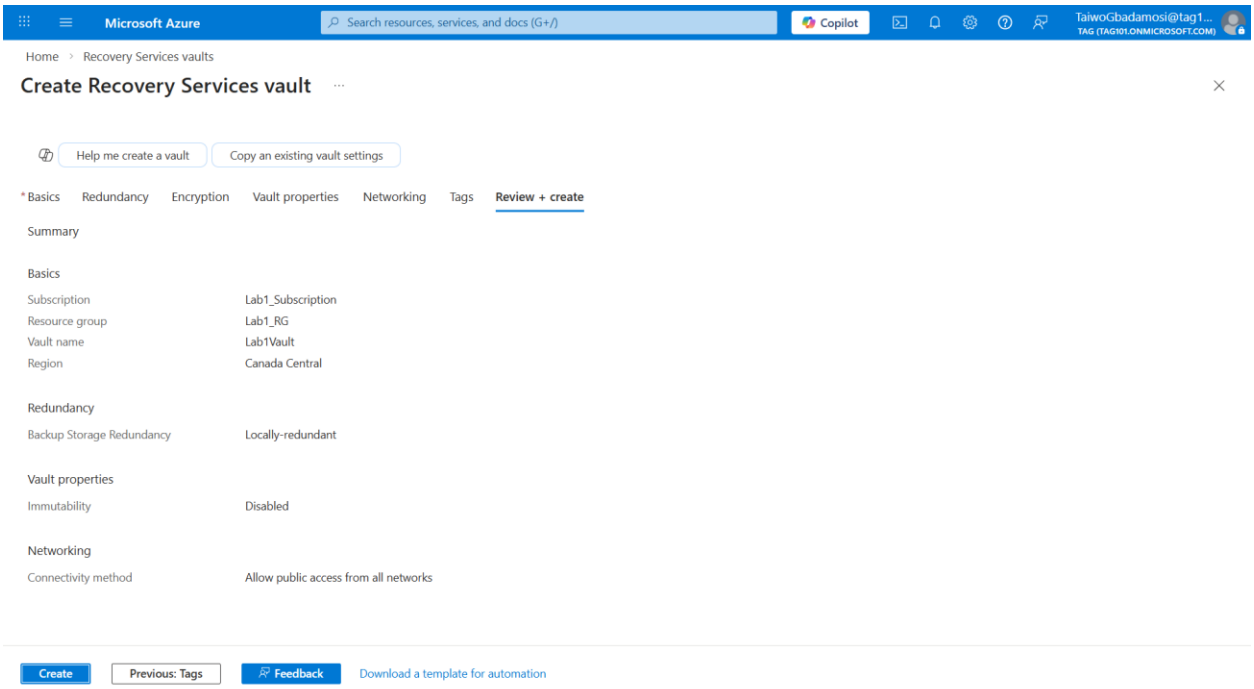
Accessing Azure File Share Subfolder "Documents" via Mapped Drive



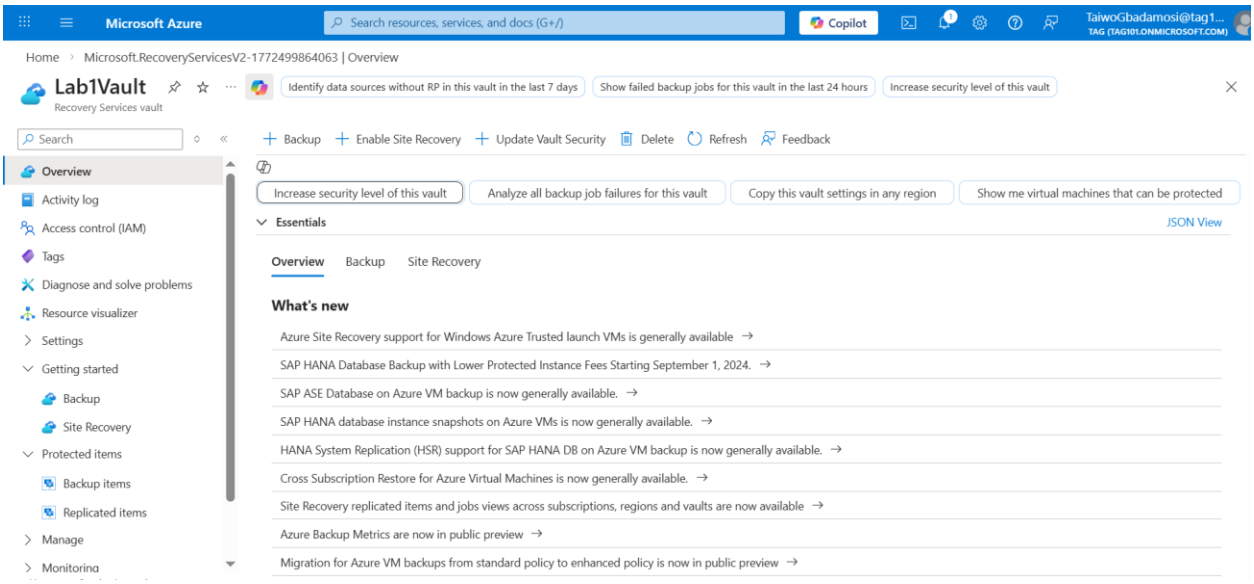
Accessing Azure File Share file within subfolder "Documents" via Mapped Drive

Phase III: Data Protection (Azure Backup)

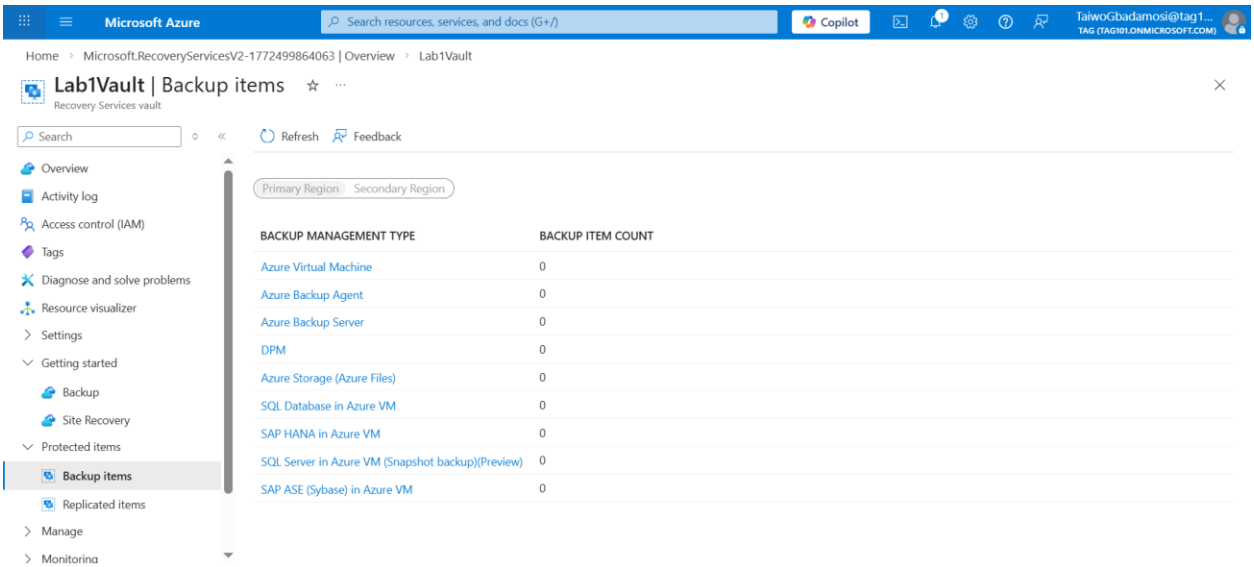
- **Recovery Services Vault (RSV):** Deployed a central vault for backup management.



Configuration of Azure Recovery services vault

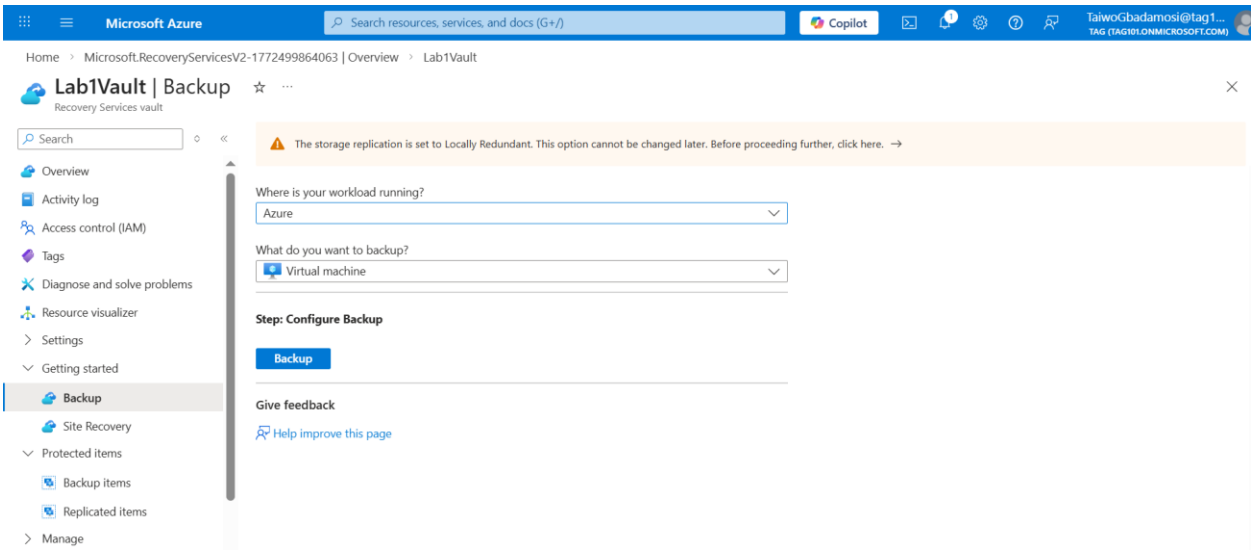


Azure recovery services vault overview

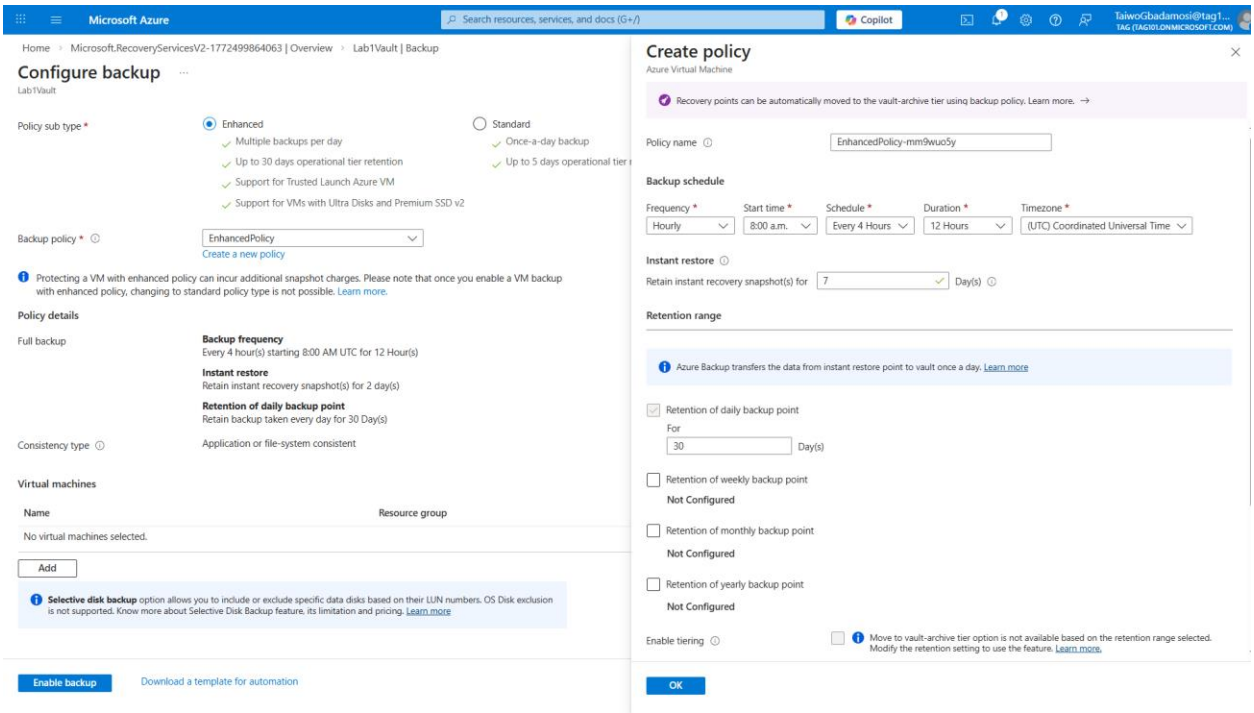


Azure recovery services vault backup management list

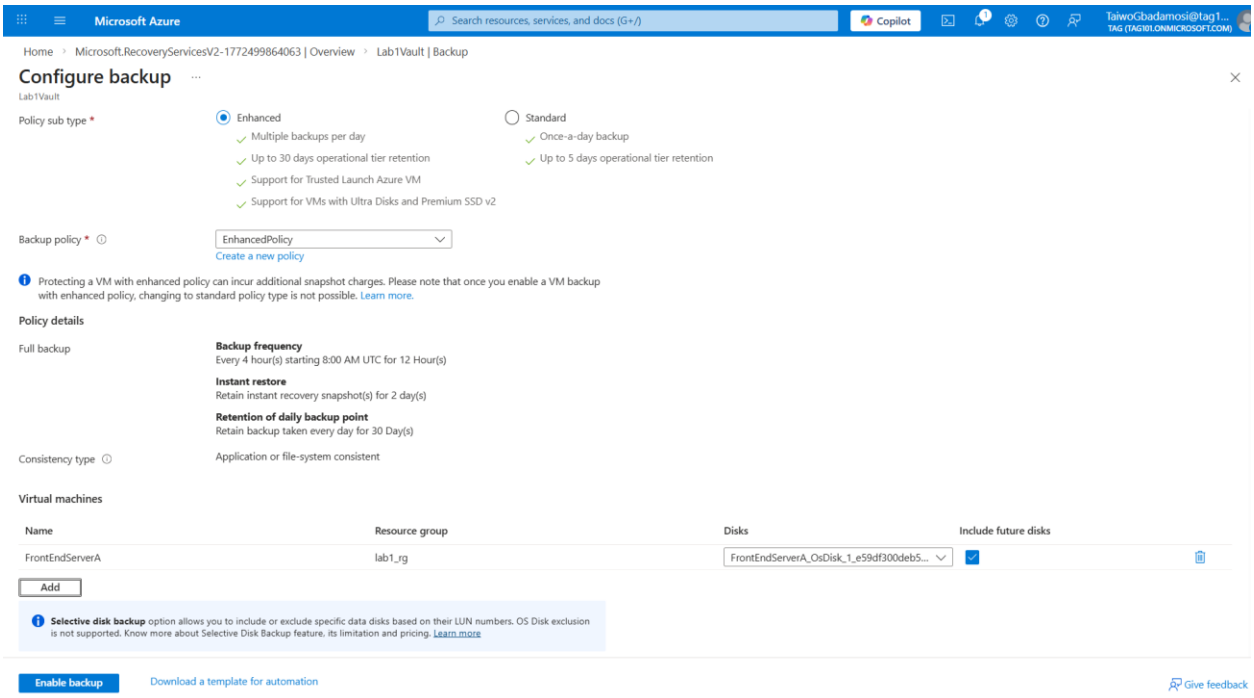
- **VM Backup:** Configured a backup policy (Frequency/Retention) for the Azure VM.



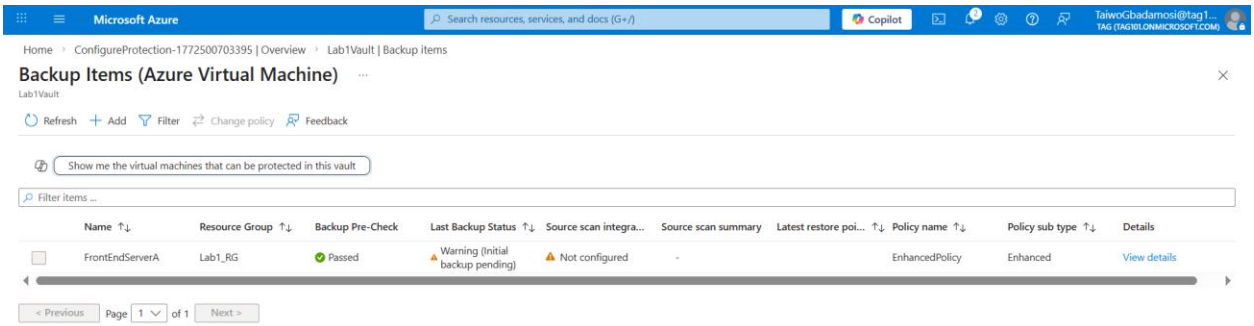
Azure virtual machine backup selection



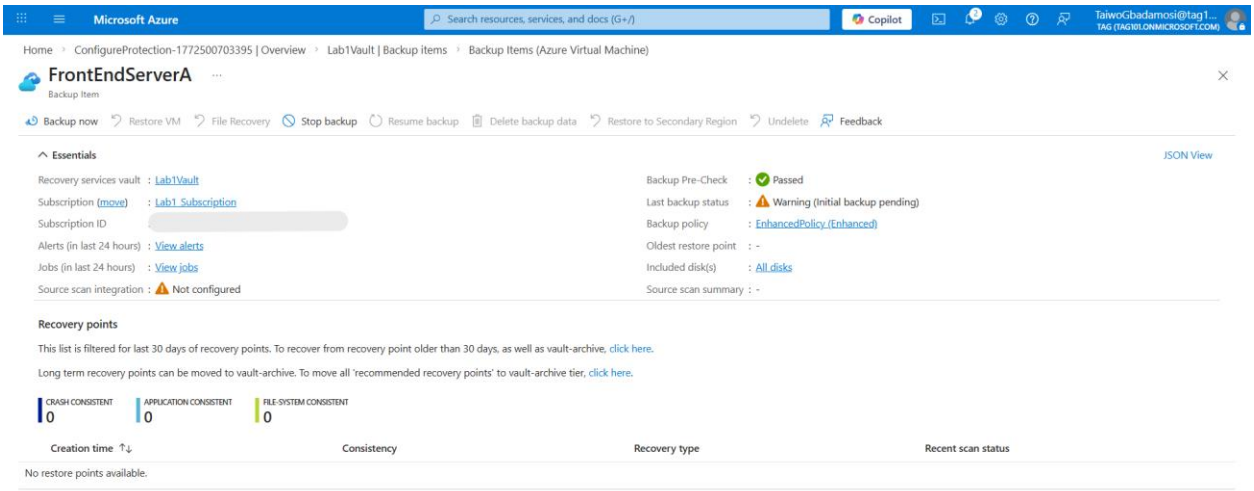
Azure virtual machine backup policy configuration



Azure virtual machine backup configuration

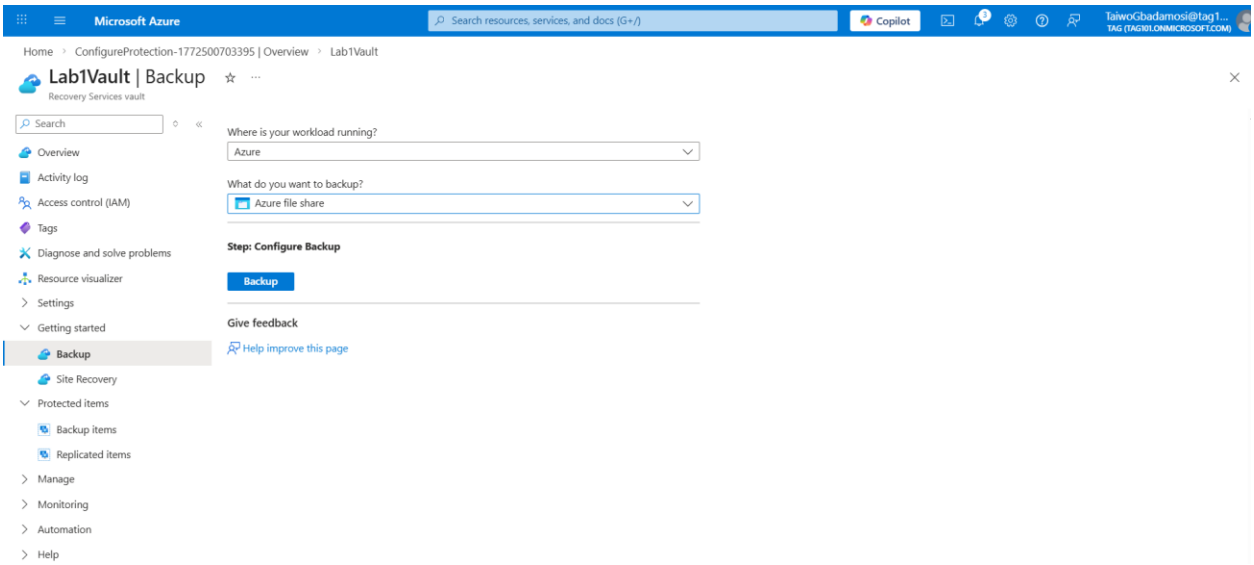


Azure virtual machine backup registered: Initial Backup Pending

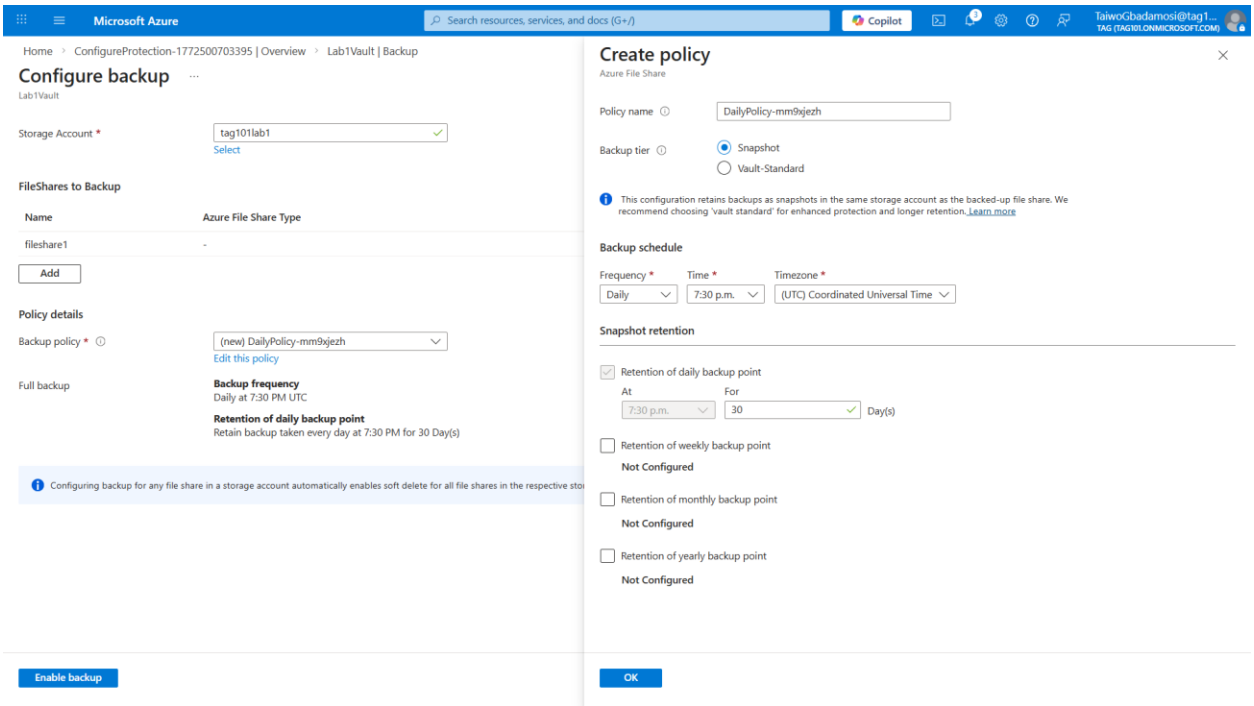


Azure virtual machine backup configuration overview

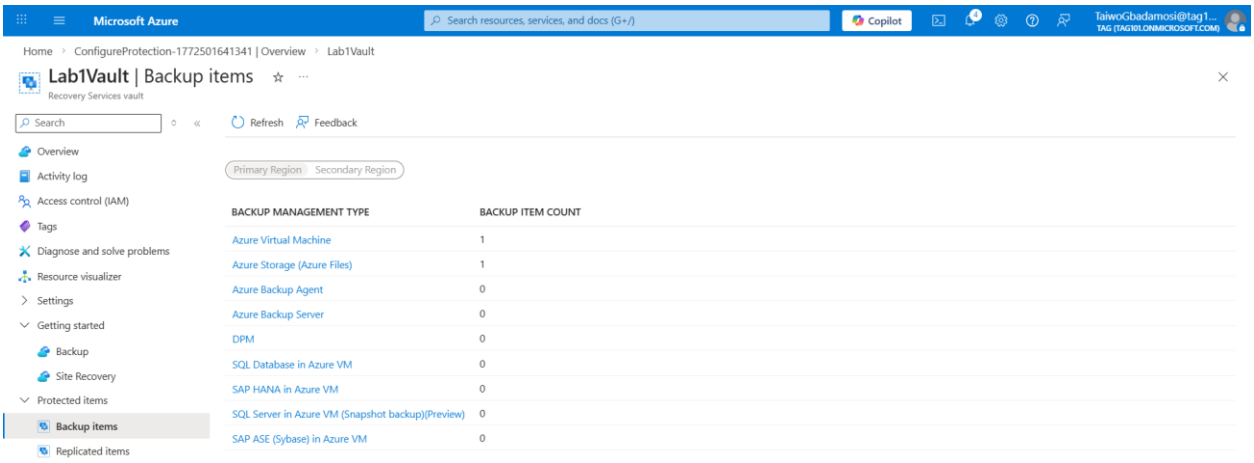
- **File Share Backup:** Enabled snapshots and backups for the Azure File Share.



Azure file share backup selection

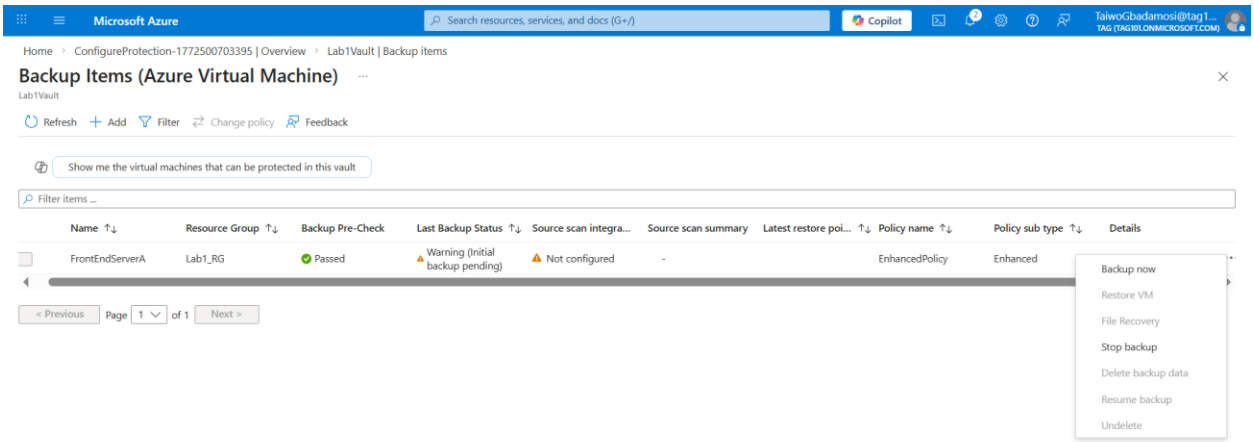


Azure file shares backup policy configuration

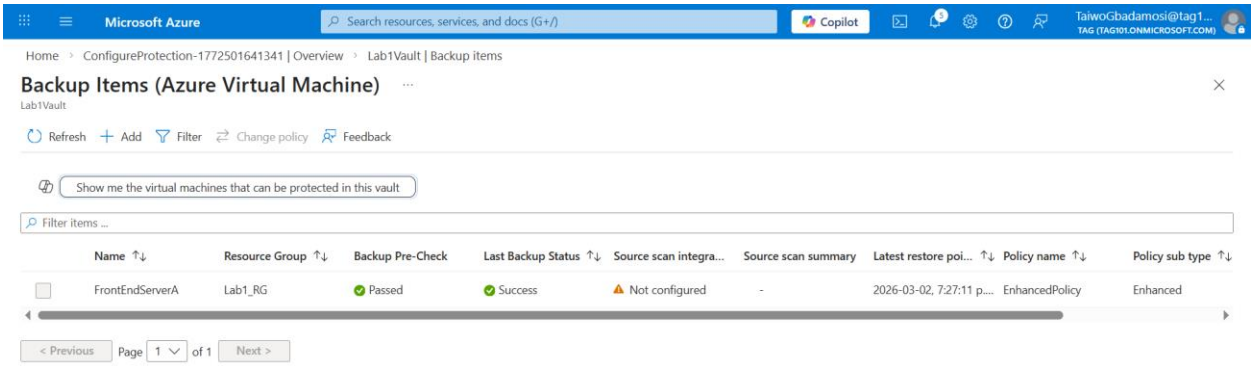


Azure backup items overview showing configured backups

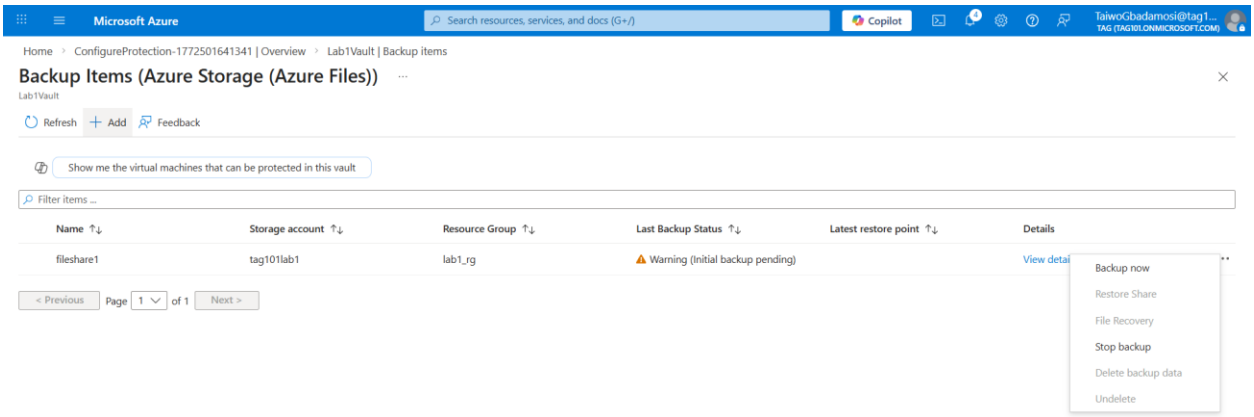
- **Manual Trigger:** Performed an "On-Demand" backup (Backup Now) to ensure immediate data protection.



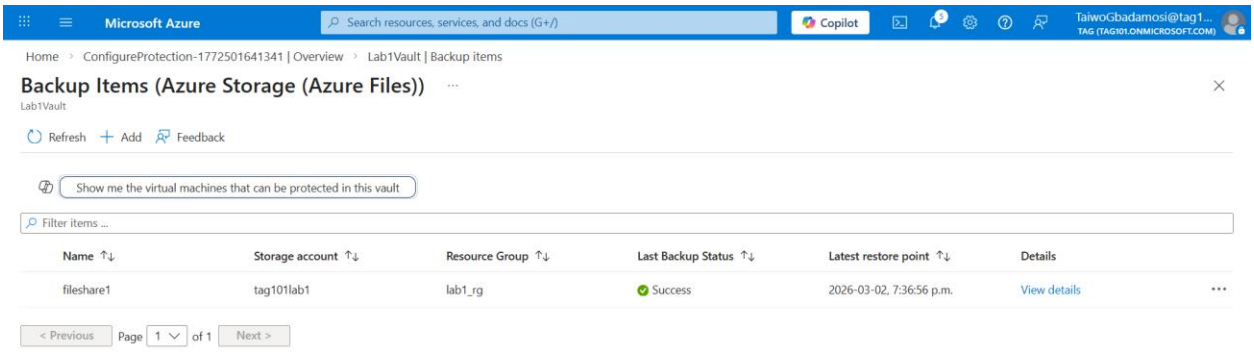
Initiating on-demand "Backup now" via Recovery Services Vault



Validation of completed backup job & point in time recovery state

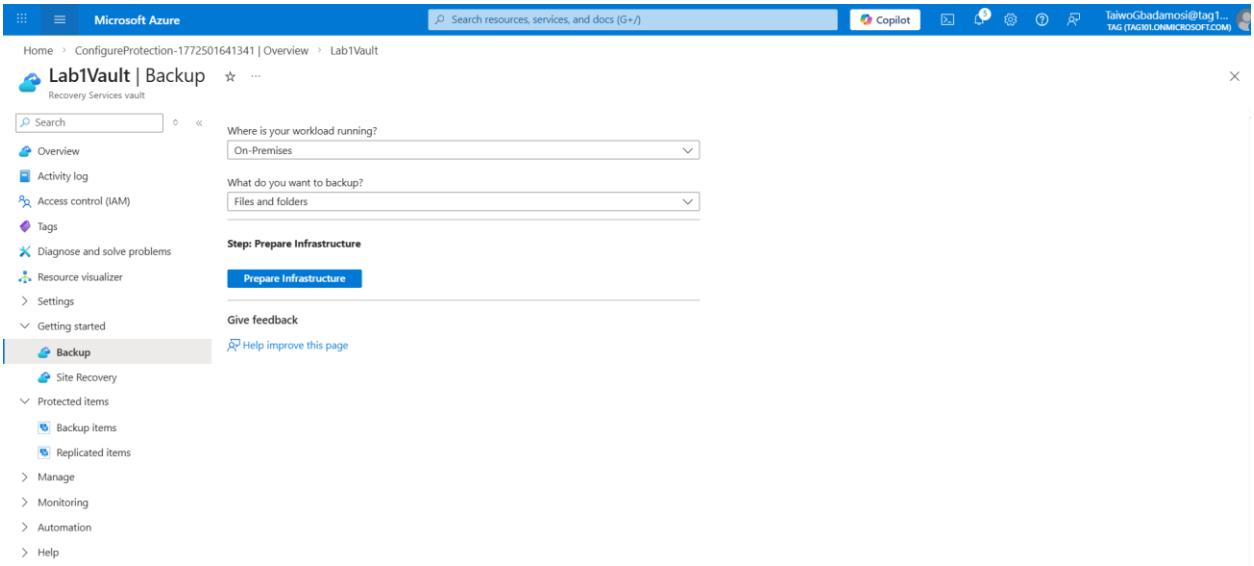


Initiating on-demand "Backup now" via Recovery Services Vault

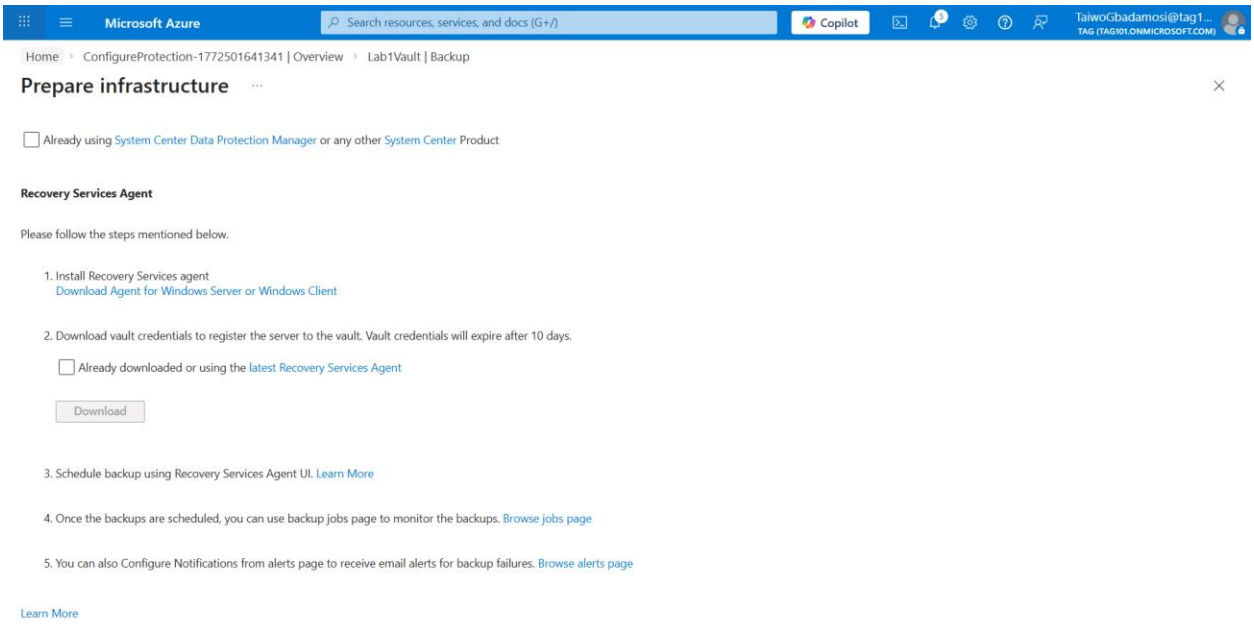


Validation of completed backup Job & point in time recovery state

- **Hybrid Readiness:** Explored the "On-Premises" backup settings, showcasing the ability to protect local servers using the same vault infrastructure.



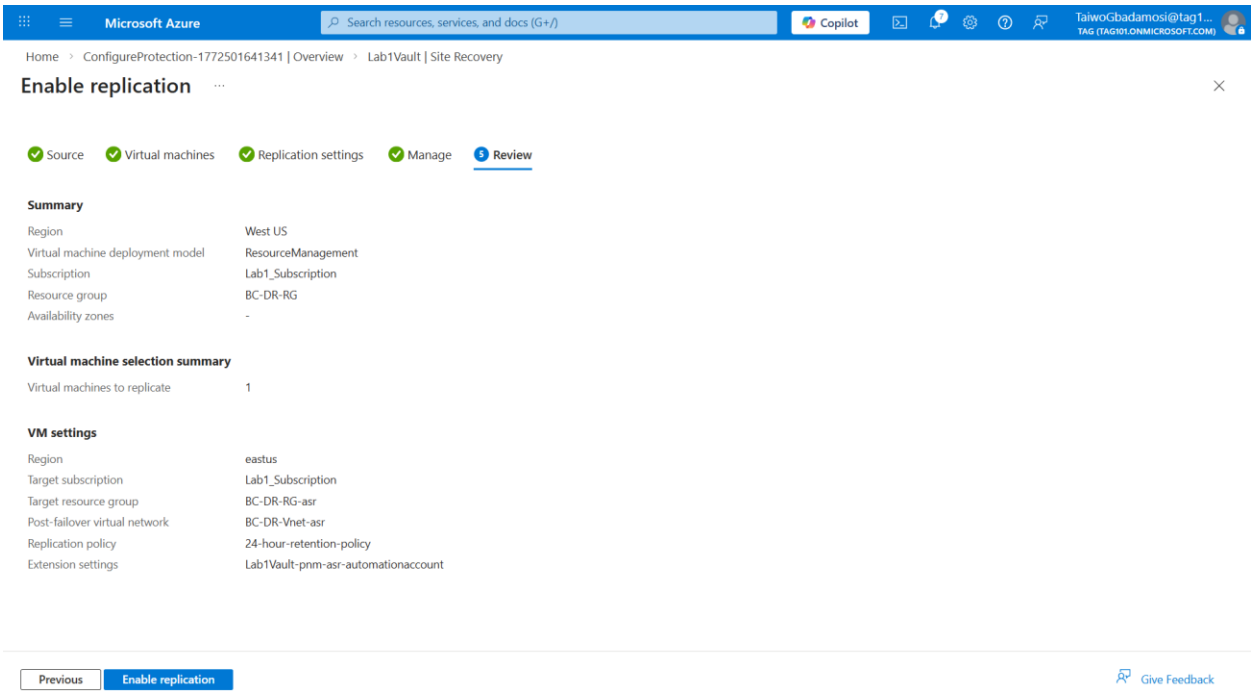
Hybrid Integration: Configuring backup on-premises assets



Hybrid Integration: Prepare infrastructure for local server protection (including steps for configuration)

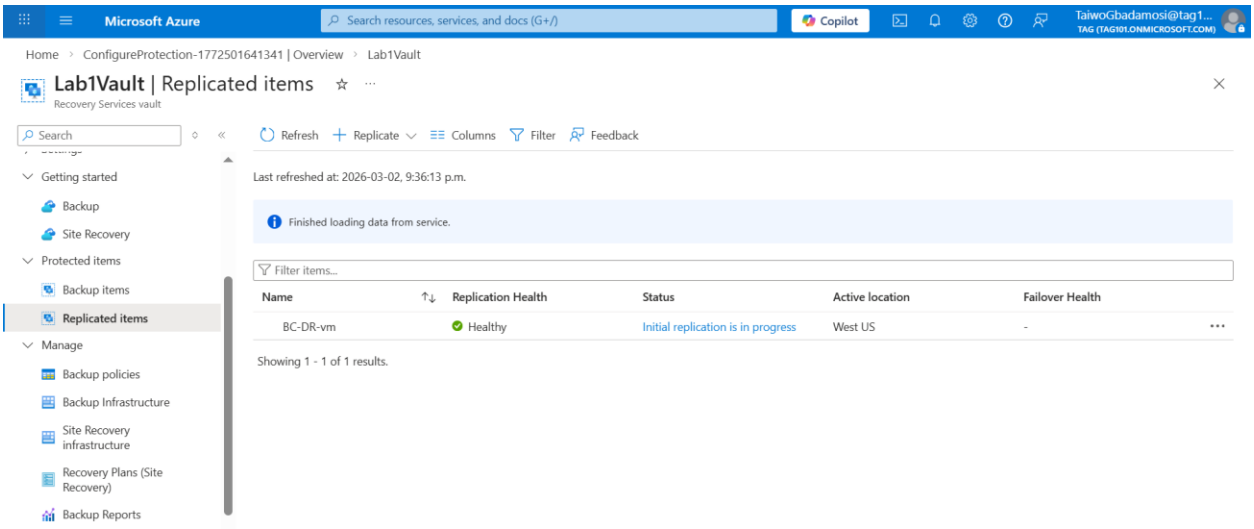
Phase IV: Disaster Recovery & Replication

- **Replication Policy:** Defined the source and target regions for resource redundancy.

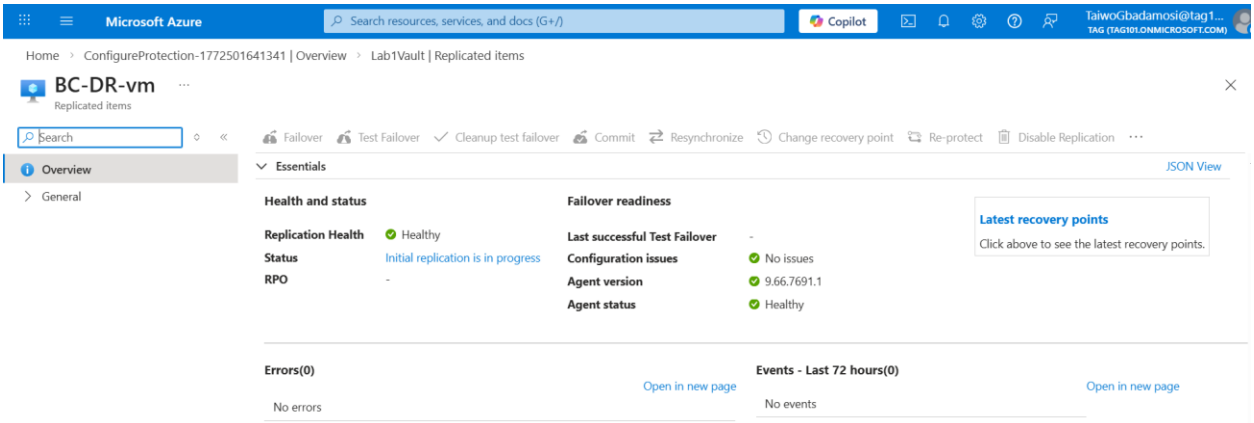


Integrating replication for Business Continuity

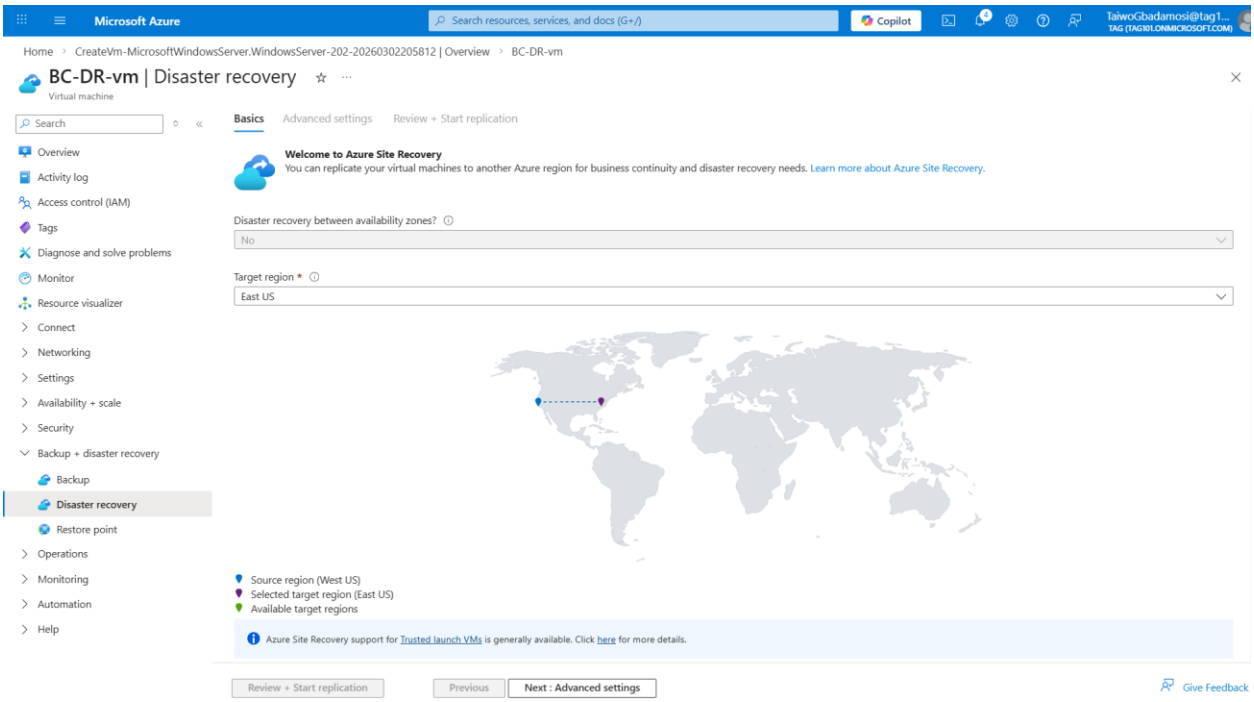
- **Azure Site Recovery (ASR):** Initiated the replication of the VM and associated data.



Initiated replication of virtual machine and associated data

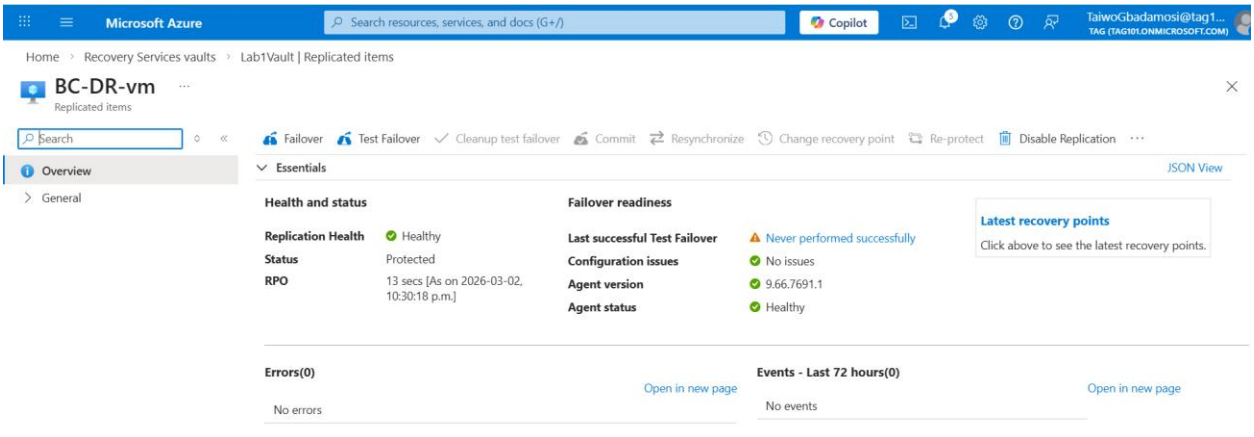


Replicated Azure virtual machine overview



Replicated Azure virtual machine disaster recovery topology map

- **BCDR Readiness:** Reviewed the Failover and Test Failover settings, preparing the environment for a regional outage scenario.



Azure virtual machine replication status with configurable failover blade

5. Summary & Lessons Learned

Summary

The project successfully demonstrates a defense-in-depth approach to data. By combining network-level security, identity-level security, and physical-level security through replication/backups, the infrastructure is resilient against both cyberattacks and regional hardware failures.

Lessons Learned

- **Network-Level Security:** Azure Storage firewalls are highly effective, even with the correct credentials, an unallowed IP is blocked at the edge.
- **Storage Tiers:** Implementing Lifecycle Management taught me how to balance high availability with cost-efficiency by moving stale data to cheaper tiers.
- **Recovery Objectives:** Backup (Restoring data from a point in time) and Replication (Switching to a live copy in another region) are two different data protection strategies but complementary pillars of Business continuity/Disaster recovery.
- **Hybrid Capabilities:** Exploring the On-Premise backup settings highlighted how Azure acts as a bridge for companies moving from local data centers to the cloud.